CORE PURE / PROOF BY INDUCTION

Proof by Induction - Divisibility

Worked solutions 8 questions · 48 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

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1. Prove by induction that

$$n^5 - n$$

is divisible by 5 for all positive integers n **[6]**

Solution

We prove by induction that $n^5 - n$ is divisible by 5 for all positive integers n.

Base case: $n = 1$. Then

$$1^5 - 1 = 1 - 1 = 0$$

This is divisible by 5, so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$k^5 - k \text{ is divisible by } 5$$

so there is an integer m such that

$$k^5 - k = 5m$$

Inductive step: We must show that $(k+1)^5 - (k+1)$ is also divisible by 5.

Let $f(n) = n^5 - n$. Then

$$f(k+1) = f(k) + \big(f(k+1) - f(k)\big)$$

Now calculate the difference:

$$\begin{aligned} f(k+1) - f(k) &= \big((k+1)^5 - (k+1)\big) - \big(k^5 - k\big) \\ &= (k+1)^5 - k^5 - 1 \\ &= \big(k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1\big) - k^5 - 1 \\ &= 5k^4 + 10k^3 + 10k^2 + 5k \\ &= 5(k^4 + 2k^3 + 2k^2 + k) \end{aligned}$$

So

$$(k+1)^5 - (k+1) = (k^5 - k) + 5(k^4 + 2k^3 + 2k^2 + k)$$

By the inductive hypothesis, $k^5 - k$ is divisible by 5. Also, $5(k^4 + 2k^3 + 2k^2 + k)$ is clearly divisible by 5.

Therefore $(k+1)^5 - (k+1)$ is divisible by 5, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all positive integers n.

2. For each positive integer n, define

$$u_n = 5^{2n} - 2^n$$

Use proof by induction to show that 23 is a factor of u_n for all $n \in \mathbb{Z}^+$ **[6]**

Solution

We prove by induction that $u_n = 5^{2n} - 2^n$ is divisible by 23 for all $n \in \mathbb{Z}^+$.

Base case: $n = 1$. Then

$$u_1 = 5^2 - 2^1 = 25 - 2 = 23$$

This is divisible by 23, so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$u_k \text{ is divisible by } 23$$

so there is an integer m such that

$$u_k = 5^{2k} - 2^k = 23m$$

Inductive step: Now consider u_{k+1}:

$$\begin{aligned} u_{k+1} &= 5^{2(k+1)} - 2^{k+1} \\ &= 5^{2k+2} - 2 \times 2^k \\ &= 25 \times 5^{2k} - 2 \times 2^k \\ &= (23 + 2)5^{2k} - 2 \times 2^k \\ &= 23 \times 5^{2k} + 2 \times 5^{2k} - 2 \times 2^k \\ &= 23 \times 5^{2k} + 2\left(5^{2k} - 2^k\right) \end{aligned}$$

Using the inductive hypothesis $5^{2k} - 2^k = 23m$:

$$\begin{aligned} u_{k+1} &= 23 \times 5^{2k} + 2(23m) \\ &= 23\left(5^{2k} + 2m\right) \end{aligned}$$

Since $5^{2k} + 2m$ is an integer, u_{k+1} is divisible by 23, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all $n \in \mathbb{Z}^+$.

3. Prove by induction that

$$9^n - 8n - 1$$

is divisible by 64 for all positive integers n **[6]**

Solution

We prove by induction that $9^n - 8n - 1$ is divisible by 64 for all positive integers n.

Base case: $n = 1$. Then

$$9^1 - 8(1) - 1 = 9 - 8 - 1 = 0$$

This is divisible by 64, so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$9^k - 8k - 1 \text{ is divisible by } 64$$

so there is an integer m such that

$$9^k - 8k - 1 = 64m$$

Inductive step: Now consider $9^{k+1} - 8(k+1) - 1$:

$$\begin{aligned} 9^{k+1} - 8(k+1) - 1 &= 9 \times 9^k - 8k - 8 - 1 \\ &= 9 \times 9^k - 8k - 9 \\ &= 9(9^k - 8k - 1) + 64k \end{aligned}$$

Now use the inductive hypothesis $9^k - 8k - 1 = 64m$:

$$\begin{aligned} 9^{k+1} - 8(k+1) - 1 &= 9(64m) + 64k \\ &= 64(9m + k) \end{aligned}$$

Since $9m + k$ is an integer, $9^{k+1} - 8(k+1) - 1$ is divisible by 64, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all positive integers n.

4. Prove by mathematical induction that, for all non-negative integers n,

$$2^n + 2 \times 3^{2n} + 4^{2n+1}$$

is divisible by 7 **[6]**

Solution

We prove by induction that $2^n + 2 \times 3^{2n} + 4^{2n+1}$ is divisible by 7 for all non-negative integers n.

Base case: $n = 0$. Then

$$2^0 + 2 \times 3^0 + 4^1 = 1 + 2 + 4 = 7$$

This is divisible by 7, so the statement is true for $n = 0$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a non-negative integer. That is, assume

$$2^k + 2 \times 3^{2k} + 4^{2k+1} \text{ is divisible by } 7$$

so there is an integer m such that

$$2^k + 2 \times 3^{2k} + 4^{2k+1} = 7m$$

Inductive step: We must show that

$$2^{k+1} + 2 \times 3^{2k+2} + 4^{2k+3}$$

is divisible by 7.

Starting with this expression:

$$\begin{aligned} 2^{k+1} + 2 \times 3^{2k+2} + 4^{2k+3} &= 2 \times 2^k + 2 \times 3^2 \times 3^{2k} + 4^2 \times 4^{2k+1} \\ &= 2 \times 2^k + 18 \times 3^{2k} + 16 \times 4^{2k+1} \\ &= 2\left(2^k + 2 \times 3^{2k} + 4^{2k+1}\right) + 14 \times 3^{2k} + 14 \times 4^{2k+1} \end{aligned}$$

Now use the inductive hypothesis $2^k + 2 \times 3^{2k} + 4^{2k+1} = 7m$:

$$\begin{aligned} 2^{k+1} + 2 \times 3^{2k+2} + 4^{2k+3} &= 2(7m) + 14 \times 3^{2k} + 14 \times 4^{2k+1} \\ &= 14m + 14\left(3^{2k} + 4^{2k+1}\right) \\ &= 7\left(2m + 2 \times 3^{2k} + 2 \times 4^{2k+1}\right) \end{aligned}$$

Since $2m + 2 \times 3^{2k} + 2 \times 4^{2k+1}$ is an integer, $2^{k+1} + 2 \times 3^{2k+2} + 4^{2k+3}$ is divisible by 7, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 0$, by the principle of mathematical induction the statement is true for all non-negative integers n.

5. Prove by induction that, for $n \in \mathbb{Z}^+$,

$$7^n - 6n - 1$$

is divisible by 36 **[6]**

Solution

We prove by induction that $7^n - 6n - 1$ is divisible by 36 for all $n \in \mathbb{Z}^+$.

Base case: $n = 1$. Then

$$7^1 - 6(1) - 1 = 7 - 6 - 1 = 0$$

This is divisible by 36, so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$7^k - 6k - 1 \text{ is divisible by } 36$$

so there is an integer m such that

$$7^k - 6k - 1 = 36m$$

Inductive step: We must show that

$$7^{k+1} - 6(k+1) - 1$$

is divisible by 36.

Starting with this expression:

$$\begin{aligned} 7^{k+1} - 6(k+1) - 1 &= 7^{k+1} - 6k - 6 - 1 \\ &= 7^{k+1} - 6k - 7 \end{aligned}$$

Now rewrite it to use the inductive hypothesis:

$$\begin{aligned} 7^{k+1} - 6k - 7 &= 7 \times 7^k - 6k - 7 \\ &= 7(7^k - 6k - 1) + 36k \end{aligned}$$

Using the inductive hypothesis $7^k - 6k - 1 = 36m$:

$$\begin{aligned} 7^{k+1} - 6(k+1) - 1 &= 7(36m) + 36k \\ &= 36(7m + k) \end{aligned}$$

Since $7m + k$ is an integer, $7^{k+1} - 6(k+1) - 1$ is divisible by 36, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all $n \in \mathbb{Z}^+$.

6. The function f is defined by

$$f(n) = 9^n - 8n + 63$$

Prove by induction that $f(n)$ is divisible by 64 for $n \geq 1$ **[6]**

Solution

We prove by induction that $f(n) = 9^n - 8n + 63$ is divisible by 64 for all $n \geq 1$.

Base case: $n = 1$. Then

$$f(1) = 9^1 - 8(1) + 63 = 9 - 8 + 63 = 64$$

This is divisible by 64, so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$f(k) = 9^k - 8k + 63 \text{ is divisible by } 64$$

so there is an integer m such that

$$9^k - 8k + 63 = 64m$$

Inductive step: Now consider $f(k+1)$:

$$\begin{aligned} f(k+1) &= 9^{k+1} - 8(k+1) + 63 \\ &= 9 \times 9^k - 8k - 8 + 63 \\ &= 9 \times 9^k - 8k + 55 \end{aligned}$$

We now rewrite this in terms of $f(k) = 9^k - 8k + 63$:

$$\begin{aligned} f(k+1) &= 9(9^k - 8k + 63) + 72k - 567 - 8k + 55 \\ &= 9(9^k - 8k + 63) + 64k - 512 \\ &= 9(9^k - 8k + 63) + 64(k - 8) \end{aligned}$$

Using the inductive hypothesis $9^k - 8k + 63 = 64m$:

$$\begin{aligned} f(k+1) &= 9(64m) + 64(k - 8) \\ &= 64(9m + k - 8) \end{aligned}$$

Since $9m + k - 8$ is an integer, $f(k+1)$ is divisible by 64, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all $n \geq 1$.

7. Prove by mathematical induction that

$$7^{2n} + 8 \times 13^n - 9$$

is divisible by 36 for all positive integers n **[6]**

Solution

Let

$$E_n = 7^{2n} + 8 \times 13^n - 9$$

We prove by induction that E_n is divisible by 36 for all positive integers n.

Base case: $n = 1$. Then

$$\begin{aligned} E_1 &= 7^2 + 8 \times 13 - 9 \\ &= 49 + 104 - 9 \\ &= 144 \\ &= 36 \times 4 \end{aligned}$$

This is divisible by 36, so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$E_k = 7^{2k} + 8 \times 13^k - 9 \text{ is divisible by } 36$$

so there is an integer m such that

$$7^{2k} + 8 \times 13^k - 9 = 36m$$

Inductive step: Consider E_{k+1}:

$$\begin{aligned} E_{k+1} &= 7^{2(k+1)} + 8 \times 13^{k+1} - 9 \\ &= 7^{2k+2} + 8 \times 13^{k+1} - 9 \\ &= 49 \times 7^{2k} + 104 \times 13^k - 9 \end{aligned}$$

Now rewrite this so that E_k appears:

$$\begin{aligned} E_{k+1} &= 13 \times 7^{2k} + 36 \times 7^{2k} + 13 \times 8 \times 13^k - 117 + 108 \\ &= 13\big(7^{2k} + 8 \times 13^k - 9\big) + 36\big(7^{2k} + 3\big) \\ &= 13E_k + 36\big(7^{2k} + 3\big) \end{aligned}$$

Using the inductive hypothesis $E_k = 36m$:

$$\begin{aligned} E_{k+1} &= 13(36m) + 36\big(7^{2k} + 3\big) \\ &= 36\big(13m + 7^{2k} + 3\big) \end{aligned}$$

Since $13m + 7^{2k} + 3$ is an integer, E_{k+1} is divisible by 36, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all positive integers n.

8. Prove by induction that, for $n \in \mathbb{Z}^+$,

$$13^n + 3 \times 4^n + 11$$

is divisible by 12 **[6]**

Solution

We prove by induction that $13^n + 3 \times 4^n + 11$ is divisible by 12 for all $n \in \mathbb{Z}^+$.

Base case: $n = 1$. Then

$$13^1 + 3 \times 4^1 + 11 = 13 + 12 + 11 = 36 = 12 \times 3$$

This is divisible by 12, so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$13^k + 3 \times 4^k + 11 \text{ is divisible by } 12$$

so there is an integer m such that

$$13^k + 3 \times 4^k + 11 = 12m$$

Inductive step: We must show that $13^{k+1} + 3 \times 4^{k+1} + 11$ is divisible by 12.

Consider the difference:

$$\begin{aligned}
&\left(13^{k+1} + 3 \times 4^{k+1} + 11\right) - \left(13^k + 3 \times 4^k + 11\right) \\
&= 13^{k+1} - 13^k + 3 \times 4^{k+1} - 3 \times 4^k \\
&= 13^k(13 - 1) + 3 \times 4^k(4 - 1) \\
&= 12 \times 13^k + 9 \times 4^k \\
&= 12 \times 13^k + 12\left(3 \times 4^{k-1}\right)
\end{aligned}$$

Since $k \geq 1$, 4^{k-1} is an integer, so this whole difference is a multiple of 12.

Therefore

$$\begin{aligned}
13^{k+1} + 3 \times 4^{k+1} + 11 &= \left(13^k + 3 \times 4^k + 11\right) + \left[\left(13^{k+1} + 3 \times 4^{k+1} + 11\right) - \left(13^k + 3 \times 4^k + 11\right)\right] \\
&= 12m + 12 \times 13^k + 12\left(3 \times 4^{k-1}\right) \\
&= 12\left(m + 13^k + 3 \times 4^{k-1}\right)
\end{aligned}$$

Since $m + 13^k + 3 \times 4^{k-1}$ is an integer, $13^{k+1} + 3 \times 4^{k+1} + 11$ is divisible by 12, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all $n \in \mathbb{Z}^+$.